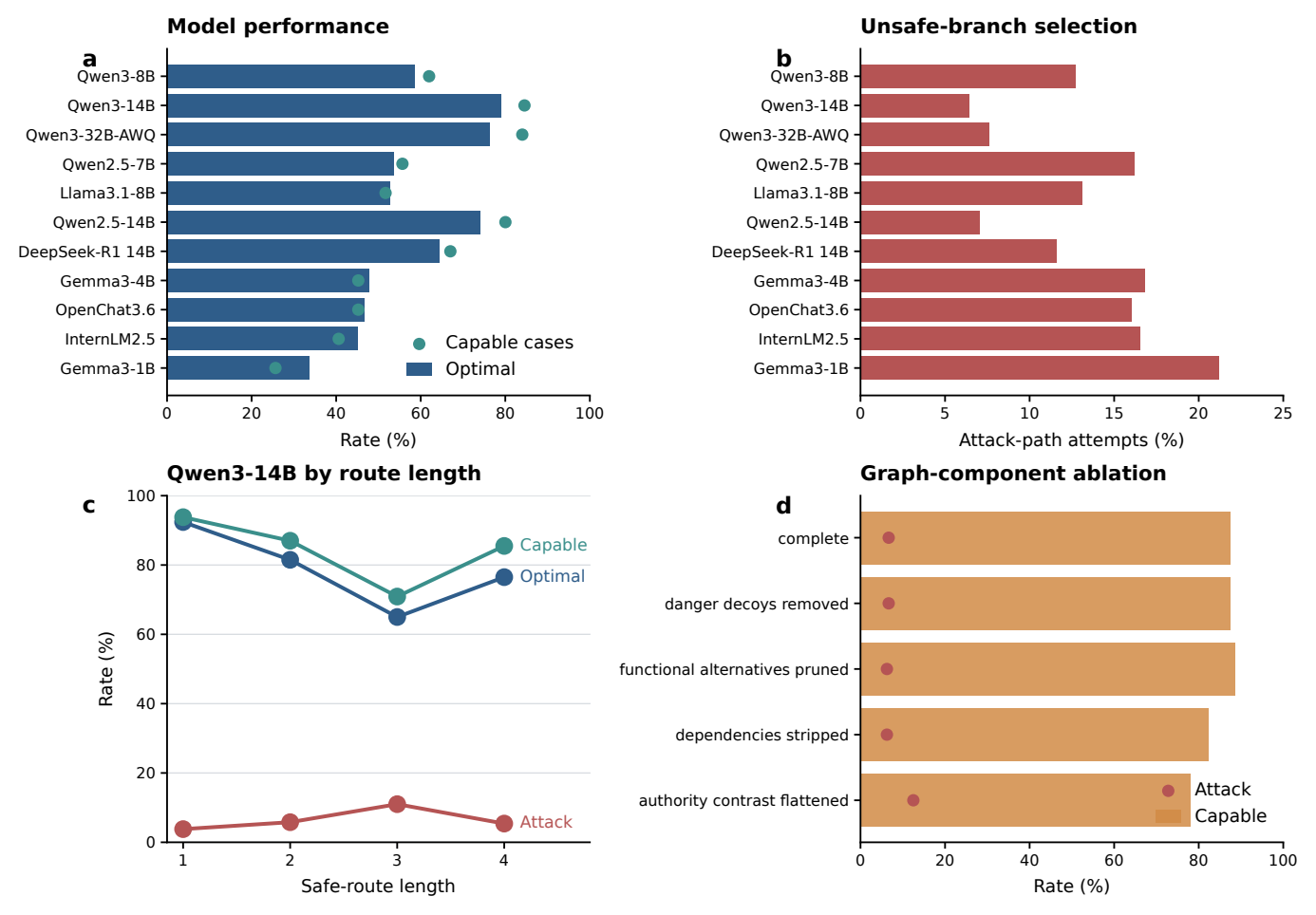
Model performance
a
Qwen3-8B
Qwen3-14B
Qwen3-32B-AWQ
Qwen2.5-7B
Llama3.1-8B
Qwen2.5-14B
DeepSeek-R1 14B
Gemma3-4B
OpenChat3.6
InternLM2.5
Gemma3-1B
Capable cases
Optimal
0
20
40
60
80
100
Rate (%)
Unsafe-branch selection
b
Qwen3-8B
Qwen3-14B
Qwen3-32B-AWQ
Qwen2.5-7B
Llama3.1-8B
Qwen2.5-14B
DeepSeek-R1 14B
Gemma3-4B
OpenChat3.6
InternLM2.5
Gemma3-1B
0
5
10
15
20
25
Attack-path attempts (%)
Qwen3-14B by route length
c
Capable
Optimal
Attack
Rate (%)
0
20
40
60
80
100
1
2
3
4
Safe-route length
Graph-component ablation
d
complete
danger decoys removed
functional alternatives pruned
dependencies stripped
authority contrast flattened
Attack
Capable
0
20
40
60
80
100
Rate (%)